Chest X-Ray Pneumonia Detection using Deep Convolutional Neural Networks

Name: Hossam Mohamed Refaat

Instructor: ENG/ Mohamed Waleed

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1. Introduction:

Pneumonia is a life-threatening inflammatory condition of the lung primarily affecting the microscopic air sacs known as alveoli. It is typically caused by infection with viruses or bacteria and requires timely and accurate diagnosis to ensure appropriate medical intervention. Chest X-ray (CXR) radiography is the most commonly used diagnostic imaging modality for detecting pneumonia due to its availability and low cost. However, manual evaluation of CXRs requires specialized clinical expertise and is susceptible to subjective interpretation and human error, particularly when differentiating between bacterial and viral etiologies.

Recent advancements in Deep Learning, particularly Convolutional Neural Networks (CNNs), have demonstrated near-human performance in medical imaging diagnostics. This project presents an end-to-end Computer-Aided Diagnosis (CAD) system designed to automate chest X-ray classification using a custom-designed Deep CNN.

2. Problem Statement:

Accurate differentiation between normal chest X-rays, bacterial pneumonia, and viral pneumonia is clinically critical because the therapeutic approaches differ fundamentally: bacterial pneumonia requires immediate antibiotic administration, whereas viral pneumonia requires supportive care or antiviral agents. Misdiagnosis can lead to inappropriate treatments, antibiotic misuse, or delayed recovery.

From a machine learning perspective, this task poses a multi-class image classification challenge characterized by high intra-class variance, low inter-class variation (visual similarities between viral and bacterial lesions), and severe class imbalance in real-world clinical datasets.

3. Objectives:

The primary objectives of this project are:

- To develop an end-to-end Deep Learning pipeline using Python and TensorFlow/Keras for multi-class pneumonia classification.

- To perform Exploratory Data Analysis (EDA) and apply targeted preprocessing techniques to handle noise and class imbalance.

- To design, build, and optimize a custom Convolutional Neural Network (CNN) tailored for grayscale chest X-ray feature extraction.

- To evaluate the trained model on unseen test data using robust clinical metrics (Precision, Recall, F1-Score, Loss, and Confusion Matrix).

- To package the final model and code into a structured, reproducible repository following industry-standard folder guidelines.

4. Dataset Description:

The dataset utilized in this project consists of chest X-ray images structured into a 3-class classification problem:

- **Source:** Chest X-Ray Images Dataset (Pneumonia) sourced from pediatric patients.

- **Total Samples:** 618 evaluation test samples (along with training and validation splits).

- **Classes:**

    - NORMAL: Clear lungs with no clinical manifestations of infection.

    - PNEUMONIA_BACTERIA: Focal lobar consolidation typical of bacterial infection.

    - PNEUMONIA_VIRAL: Diffuse interstitial patterns typical of viral infection.

- **Directory Structure:**
    - dataset/train/
    - dataset/valid/
    - dataset/test/

5. Data Preprocessing:

To prepare the raw X-ray images for training and prevent overfitting, the following pipeline was executed:

- **Directory Normalization & Split Balancing:** Standardized folder naming (converting PNEUMONIA_BACTERIAL to PNEUMONIA_BACTERIA) and rebalanced the validation set by moving a stratified 15% sample from the training set.
- **Color Space & Resizing:** Converted all images to single-channel Grayscale (color_mode='grayscale') and resized them to 256x256 pixels to lower memory overhead while retaining diagnostic details.
- **Pixel Normalization:** Scaled pixel intensities from [0, 255] to [0, 1] using rescale=1./255.
- **Data Augmentation:** Applied real-time augmentations via ImageDataGenerator on training samples to enhance model generalization:
    - Rotation Range: +/- 15 degrees
    - Zoom Range: 10%
    - Width & Height Shift Ranges: 10%

    - Fill Mode: nearest

6. CNN Architecture:

A custom Deep Convolutional Neural Network was built specifically for single-channel X-ray classification. The architecture balances expressive capacity with computational efficiency:

- **Input Layer:** Grayscale input tensor of shape (256, 256, 1).
- **Convolutional Layers:** Multiple stacked Conv2D layers using 3x3 kernels paired with ReLU activations to extract hierarchical spatial features (edges, textures, and opacities).
- **Pooling Layers:** MaxPooling2D (2x2) following convolution blocks to progressively reduce spatial dimensions and achieve translation invariance.
- **Regularization:** Dropout layers inserted to prevent feature co-adaptation and combat overfitting.
- **Classification Head:** A Flatten layer followed by a Dense hidden layer, leading to an output Dense layer with 3 neurons activated by the Softmax function to output class probability distributions.

7. Training Process:

The model was configured and trained under the following hyperparameter settings:

- **Loss Function:** categorical_crossentropy
- **Optimizer:** Adam optimizer with an initial learning rate of 0.001.
- **Batch Size:** 32
- **Max Epochs:** 15 epochs

- **Callbacks:**
    - **Early Stopping:** Configured to monitor val_loss with a patience=5 and restore_best_weights=True to halt training when validation convergence was reached.
- **Class Weighting:** Implemented compute_class_weight('balanced') to assign higher penalties to minority class errors (PNEUMONIA_VIRAL), offsetting class imbalance.

8. Evaluation:

The model was evaluated on an unseen held-out test set (N = 618 samples). Performance was evaluated using Accuracy, Loss, Precision, Recall, and F1-Score:

Overall Metrics

- **Test Accuracy:** 85.44%
- **Test Loss:** 0.5114
- **Macro Average F1-Score:** 0.84
- **Weighted Average F1-Score:** 0.85

Classification Report Summary

Class	Precision	Recall	F1-Score	Support
NORMAL	**0.96**	0.79	0.87	231
PNEUMONIA_BACTERIA	0.82	**0.96**	**0.88**	240
PNEUMONIA_VIRAL	0.78	0.78	0.78	147

9. Results:

- **Bacterial Pneumonia Performance:** The network achieved exceptionally strong sensitivity for bacterial cases, yielding a Recall of 0.96 (230 out of 240 correct diagnoses). This is clinically advantageous as missing a bacterial infection carries severe health risks.
- **Normal Case Specificity:** High precision of 0.96 for healthy controls ensures minimal false alarms for non-diseased individuals.
- **Visualizations Included in Submission:**
  - Exploratory Data Distribution Plots (Bar plot & Pie chart for training class proportions).
  - Sample X-ray Grid showing normalized visual representations.
  - Confusion Matrix highlighting inter-class misclassifications.

10. Challenges:

- **Inter-Class Visual Ambiguity:** Distinguishing between viral pneumonia and bacterial pneumonia presented a significant bottleneck due to overlapping radiographic features (e.g., subtle infiltrates).
- **Class Imbalance:** The training set contained substantially fewer viral pneumonia cases compared to bacterial and normal cases. This was successfully mitigated using balanced class weighting (class_weight='balanced').

- **Hardware & Memory Constraints:** Processing high-resolution medical images required batching optimization and grayscale conversion to remain within local GPU memory limits.

11. Future Improvements:

- **Transfer Learning:** Incorporating state-of-the-art pre-trained architectures (e.g., DenseNet-121, EfficientNet, or ResNet-50) fine-tuned on medical domain data.

- **Explainable AI (XAI):** Implementing Grad-CAM (Gradient-weighted Class Activation Mapping) to visualize spatial heatmaps, allowing clinicians to verify which regions of interest influenced the CNN decisions.

- **Higher Resolution & Contrast Enhancement:** Utilizing advanced preprocessing techniques like CLAHE (Contrast Limited Adaptive Histogram Equalization) to accentuate subtle pulmonary opacities.

12. Conclusion:

In this project, a Deep Convolutional Neural Network was successfully developed, trained, and evaluated for automated 3-class pneumonia identification from chest X-rays. Achieving an overall test accuracy of 85.44% and a macro F1-score of 0.84, the system proved particularly robust in identifying bacterial pneumonia (Recall = 0.96) and normal controls (Precision = 0.96). The integration of balanced class weighting and systematic data preprocessing ensured reliable performance across all categories, proving the viability of CNNs as reliable assistive tools in clinical radiologic workflows.

13. References:

Kermany, D. S., Goldbaum, M., Cai, W., et al. (2018). *Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning.*